

Functional Constraints

Flame competition constraint within pocket regions.

Penalty coefficient 0.0100

Maximum allowable heat flux density ratio (max/min) 1000.00

The linear burning rate of the first conditional fuel is greater than that of the second.

Penalty coefficient 0.0100

Constraint on maximum kinetic flame heat flux density.

Penalty coefficient 0.0100

Maximum kinetic flame heat flux density in inter-pocket medium 1.00e+09

Maximum kinetic flame heat flux density in skeleton structure 1.00e+08

Maximum kinetic flame heat flux density in pocket region outside skeleton 1.00e+08

Constraint on skeleton thickness to pore diameter ratio.

Penalty coefficient 0.0100

Threshold for skeleton thickness to pore diameter ratio 3.00

Constraint on skeleton thickness to large oxidizer particle diameter ratio.

Penalty coefficient 0.0100

Threshold for skeleton thickness to large oxidizer particle diameter ratio 1.00

Parametric Constraints

Constraint on allowable surface temperature range

Minimum surface temperature 600 K

Maximum surface temperature 900 K

Differential Evolution Algorithm

Population size: 444

Mutation force: 0.5

Crossover probability: 0.9

Number of threads: 13

Termination strategy: OrTerminationStrategy

Group Combustion Solver Parameters

Shared Parameters (All Compositions)

Pre-exponential coefficient for gas phase reactions in inter-pocket medium 61940546183.48662

Parameter bounds [1.000E+005; 1.000E+013]

Activation energy for gas phase reactions in inter-pocket medium 223452.9708963056

Parameter bounds [5.000E+004; 2.500E+005]

Pre-exponential factor for gas-phase reactions in pocket region outside skeleton 137605.26563741174

Parameter bounds [1.000E+005; 1.000E+013]

Activation energy for gas-phase reactions in pocket region outside skeleton 50000.01330618926

Parameter bounds [5.000E+004; 2.500E+005]

Reaction order for gas-phase chemistry in inter-pocket medium 2.374993173635289

Parameter bounds [0.000E+000; 2.500E+000]

Reaction order for gas-phase chemistry outside skeleton structure 1.2682909043122645

Parameter bounds [0.000E+000; 2.500E+000]

Enthalpy difference between vaporization and binder decomposition 417198.4813639784

Parameter bounds [-1.000E+007; 1.000E+007]

Diffusion flame height correlation coefficient 1.3451

Parameter bounds [1.000E-003; 1.000E+001]

Power law exponent for burning rate dependency on A (skeleton thickness) 0.7605

Parameter bounds [0.000E+000; 3.000E+000]

Power law exponent for burning rate dependency on B (pore diameter) 0.5240

Parameter bounds [0.000E+000; 3.000E+000]

Radiation temperature correlation coefficient (average layer temperature) 0.0000

Parameter bounds [0.000E+000; 1.000E+000]

Diffusion-flame standoff pressure-factor C_{rxn} 0.00122183

Parameter bounds [0.000E+000; 5.000E+001]

Diffusion-flame standoff size-exponent m 1.98154

Parameter bounds [0.000E+000; 2.000E+000]

Bimodal-packing heat-feedback K_{pack} 9.34083

Parameter bounds [0.000E+000; 1.000E+001]
Diffusion-flame standoff pressure-exponent n_p 3.68392
Parameter bounds [2.000E+000; 4.000E+000]
WSB condensed-phase reaction K_wsb 0.199551
Parameter bounds [0.000E+000; 5.000E+000]

Bas_2 (includes Bas_3, Bas_4) - Condensed Phase Parameters

Pre-exponential coefficient for condensed phase reactions 9999692768504.07
Parameter bounds [1.000E+000; 1.000E+013]
Activation energy for condensed phase reactions 184842.19184729294
Parameter bounds [5.000E+004; 3.000E+005]
Pre-exponential factor for gas-phase reactions within skeleton structure 699732.9406957225
Parameter bounds [1.000E+005; 1.000E+013]
Activation energy for gas-phase reactions within skeleton structure 77094.29964265149
Parameter bounds [5.000E+004; 2.500E+005]
Reaction order for gas-phase chemistry in skeleton structure 0.1656864851446039
Parameter bounds [0.000E+000; 2.500E+000]
Coefficient A for metal combustion heat flux density in skeleton structure, 1.4887041844707544E-06
Parameter bounds [1.000E-010; 1.000E-003]
Coefficient B for metal combustion heat flux density in skeleton structure, 1.0699200043643123E-11
Parameter bounds [1.000E-012; 1.000E-005]

Bas_1 - Condensed Phase Parameters

Pre-exponential coefficient for condensed phase reactions 9409846821523.031
Parameter bounds [1.000E+000; 1.000E+013]
Activation energy for condensed phase reactions 136476.48451317378
Parameter bounds [5.000E+004; 3.000E+005]
Pre-exponential factor for gas-phase reactions within skeleton structure 1193994957.7606127
Parameter bounds [1.000E+005; 1.000E+013]
Activation energy for gas-phase reactions within skeleton structure 85648.18092636958
Parameter bounds [5.000E+004; 2.500E+005]
Reaction order for gas-phase chemistry in skeleton structure 1.4720260961544052
Parameter bounds [0.000E+000; 2.500E+000]
Coefficient A for metal combustion heat flux density in skeleton structure, 3.253938975547994E-06
Parameter bounds [1.000E-010; 1.000E-003]
Coefficient B for metal combustion heat flux density in skeleton structure, 1.480373333587622E-10
Parameter bounds [1.000E-012; 1.000E-005]

Bas_0 - Condensed Phase Parameters

Pre-exponential coefficient for condensed phase reactions 9999983699297.936
Parameter bounds [1.000E+000; 1.000E+013]
Activation energy for condensed phase reactions 184842.4193336069
Parameter bounds [5.000E+004; 3.000E+005]
Pre-exponential factor for gas-phase reactions within skeleton structure 197534384.80506736
Parameter bounds [1.000E+005; 1.000E+013]
Activation energy for gas-phase reactions within skeleton structure 54030.5274519502
Parameter bounds [5.000E+004; 2.500E+005]
Reaction order for gas-phase chemistry in skeleton structure 0.5132797234426084
Parameter bounds [0.000E+000; 2.500E+000]
Coefficient A for metal combustion heat flux density in skeleton structure, 1.7288162474510197E-06
Parameter bounds [1.000E-010; 1.000E-003]
Coefficient B for metal combustion heat flux density in skeleton structure, 1.4438969653482838E-06
Parameter bounds [1.000E-012; 1.000E-005]

Optimization Results

Objective function value 0.0171
Total penalty from constraint violations 0.0000

Propellant Composition "Bas_2"

At pressure 1 MPa

Experimental burning rate 16.94 mm/s
Calculated burning rate 15.83 mm/s

First Conditional Fuel

Surface temperature 825.11 K
Linear burning rate 11.99 mm/s
Mass decomposition rate $19.88 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $2.4\text{e}+07 \text{ W/m}^2$
Kinetic flame height 12 μm
Average kinetic flame density 1.75 kg/m^3
Average kinetic flame temperature 2,266.05 K
Binder sublimation heat flux density $2.4\text{e}+07 \text{ W/m}^2$
Surface heat balance error $-1.4\text{e}-04 \text{ W/m}^2$

Secondary Conditional Fuel

Surface temperature 814.38 K
Linear burning rate 8.41 mm/s
Mass decomposition rate $13.95 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $111,119.8 \text{ W/m}^2$
Kinetic flame height 1,395.44 μm
Average kinetic flame density 2.5 kg/m^3
Average kinetic flame temperature 1,589.69 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $9.14\text{e}+06 \text{ W/m}^2$
Kinetic flame height 8.5 μm
Average kinetic flame density 3.3 kg/m^3
Average kinetic flame temperature 1,202.69 K
Heat flux density due to metal combustion, $3.2\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $0.43 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $0.028 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-4.092294297874943\text{E}-08$
Skeleton layer thickness $6.55\text{e}-05 \text{ m}$
Pore diameter $1.77\text{e}-05 \text{ m}$
Diffusion flame heat flux density $241,299.64 \text{ W/m}^2$
Diffusion flame height 1,156.08 μm
Heat flux density to surface within skeletal structure $4.43\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $71,235.29 \text{ W/m}^2$
Total heat flux density to surface $1.48\text{e}+07 \text{ W/m}^2$
Binder sublimation heat flux density $1.48\text{e}+07 \text{ W/m}^2$
Surface heat balance error $-7.42\text{e}-04 \text{ W/m}^2$

At pressure 4.06 MPa

Experimental burning rate 35.09 mm/s
Calculated burning rate 36.28 mm/s

First Conditional Fuel

Surface temperature 879.86 K
Linear burning rate 64.1 mm/s
Mass decomposition rate $106.33 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $1.37\text{e}+08 \text{ W/m}^2$
Kinetic flame height 2.06 μm
Average kinetic flame density 7.02 kg/m^3
Average kinetic flame temperature 2,293.43 K
Binder sublimation heat flux density $1.37\text{e}+08 \text{ W/m}^2$
Surface heat balance error -0.015 W/m^2

Secondary Conditional Fuel

Surface temperature 834.9 K

Linear burning rate 16.44 mm/s
 Mass decomposition rate $27.28 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Region Outside Skeleton Structure
 Kinetic flame heat flux density to surface 336,371.76 W/m²
 Kinetic flame height 454.88 μm
 Average kinetic flame density 10.06 kg/m^3
 Average kinetic flame temperature 1,599.95 K
Skeleton Structure Region
 Kinetic flame heat flux density to surface $9.73\text{e}+06 \text{ W/m}^2$
 Kinetic flame height 7.77 μm
 Average kinetic flame density 13.27 kg/m^3
 Average kinetic flame temperature 1,212.95 K
 Heat flux density due to metal combustion, $5.01\text{e}+06 \text{ W/m}^2$
 Effective thermal conductivity $0.42 \text{ W/(m}\cdot\text{K)}$
 Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
 Radiative thermal conductivity $0.02 \text{ W/(m}\cdot\text{K)}$
 Conductive thermal conductivity balance error -4.092294297874943E-08
 Skeleton layer thickness $3.93\text{e}-05 \text{ m}$
 Pore diameter $1.24\text{e}-05 \text{ m}$
 Diffusion flame heat flux density $188,791.26 \text{ W/m}^2$
 Diffusion flame height 1,466.75 μm
 Heat flux density to surface within skeletal structure $4.87\text{e}+06 \text{ W/m}^2$
 Heat flux density to surface outside skeletal structure $225,252.6 \text{ W/m}^2$
 Total heat flux density to surface $1.65\text{e}+07 \text{ W/m}^2$
 Binder sublimation heat flux density $1.65\text{e}+07 \text{ W/m}^2$
 Surface heat balance error 0.0016 W/m^2

At pressure 6.5 MPa

Experimental burning rate 44.85 mm/s
 Calculated burning rate 42.84 mm/s

First Conditional Fuel

Surface temperature 900 K
 Linear burning rate 112.82 mm/s
 Mass decomposition rate $187.17 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
 Kinetic flame heat flux density to surface $2.47\text{e}+08 \text{ W/m}^2$
 Kinetic flame height 1.14 μm
 Average kinetic flame density 11.2 kg/m^3
 Average kinetic flame temperature 2,303.5 K
 Binder sublimation heat flux density $2.47\text{e}+08 \text{ W/m}^2$
 Surface heat balance error -0.044 W/m^2

Secondary Conditional Fuel

Surface temperature 839.02 K
 Linear burning rate 18.74 mm/s
 Mass decomposition rate $31.09 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

 Kinetic flame heat flux density to surface 537,172.4 W/m²
 Kinetic flame height 284.08 μm
 Average kinetic flame density 16.1 kg/m^3
 Average kinetic flame temperature 1,602.01 K

Skeleton Structure Region

 Kinetic flame heat flux density to surface $1.09\text{e}+07 \text{ W/m}^2$
 Kinetic flame height 6.9 μm
 Average kinetic flame density 21.23 kg/m^3
 Average kinetic flame temperature 1,215.01 K
 Heat flux density due to metal combustion, $5.47\text{e}+06 \text{ W/m}^2$
 Effective thermal conductivity $0.42 \text{ W/(m}\cdot\text{K)}$
 Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
 Radiative thermal conductivity $0.019 \text{ W/(m}\cdot\text{K)}$
 Conductive thermal conductivity balance error -4.092294297874943E-08

Skeleton layer thickness 3.56×10^{-5} m
Pore diameter 1.16×10^{-5} m
Diffusion flame heat flux density $166,181.49 \text{ W/m}^2$
Diffusion flame height $1,663.83 \text{ }\mu\text{m}$
Heat flux density to surface within skeletal structure $4.45 \times 10^6 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $391,233.54 \text{ W/m}^2$
Total heat flux density to surface $1.56 \times 10^7 \text{ W/m}^2$
Binder sublimation heat flux density $1.56 \times 10^7 \text{ W/m}^2$
Surface heat balance error -0.0031 W/m^2

Propellant Composition "Bas_3"

At pressure 1 MPa

Experimental burning rate 15.24 mm/s
Calculated burning rate 16 mm/s

First Conditional Fuel

Surface temperature 825.11 K
Linear burning rate 11.99 mm/s
Mass decomposition rate $19.88 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $2.4 \times 10^7 \text{ W/m}^2$
Kinetic flame height $12 \text{ }\mu\text{m}$
Average kinetic flame density 1.75 kg/m^3
Average kinetic flame temperature $2,266.05 \text{ K}$
Binder sublimation heat flux density $2.4 \times 10^7 \text{ W/m}^2$
Surface heat balance error $-1.4 \times 10^{-4} \text{ W/m}^2$

Secondary Conditional Fuel

Surface temperature 825.11 K
Linear burning rate 11.99 mm/s
Mass decomposition rate $19.88 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $78,048.98 \text{ W/m}^2$
Kinetic flame height $1,972.98 \text{ }\mu\text{m}$
Average kinetic flame density 2.49 kg/m^3
Average kinetic flame temperature $1,595.05 \text{ K}$

Skeleton Structure Region

Kinetic flame heat flux density to surface $6.32 \times 10^7 \text{ W/m}^2$
Kinetic flame height $2.84 \text{ }\mu\text{m}$
Average kinetic flame density 2.3 kg/m^3
Average kinetic flame temperature $1,724.05 \text{ K}$
Heat flux density due to metal combustion, $3.49 \times 10^6 \text{ W/m}^2$
Effective thermal conductivity $0.37 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.22 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $0.15 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-5.5152085010057306 \times 10^{-8}$
Skeleton layer thickness $5 \times 10^{-5} \text{ m}$
Pore diameter $1.47 \times 10^{-5} \text{ m}$
Diffusion flame heat flux density $478,243.65 \text{ W/m}^2$
Diffusion flame height $595.7 \text{ }\mu\text{m}$
Heat flux density to surface within skeletal structure $2.09 \times 10^7 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $53,626.64 \text{ W/m}^2$
Total heat flux density to surface $2.14 \times 10^7 \text{ W/m}^2$
Binder sublimation heat flux density $2.14 \times 10^7 \text{ W/m}^2$
Surface heat balance error -0.003 W/m^2

At pressure 4.06 MPa

Experimental burning rate 37.34 mm/s
Calculated burning rate 37.71 mm/s

First Conditional Fuel

Surface temperature 879.86 K

Linear burning rate 64.1 mm/s
Mass decomposition rate $106.33 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $1.37\text{e}+08 \text{ W/m}^2$
Kinetic flame height $2.06 \mu\text{m}$
Average kinetic flame density 7.02 kg/m^3
Average kinetic flame temperature 2,293.43 K
Binder sublimation heat flux density $1.37\text{e}+08 \text{ W/m}^2$
Surface heat balance error -0.015 W/m^2

Secondary Conditional Fuel

Surface temperature 840.88 K
Linear burning rate 19.87 mm/s
Mass decomposition rate $32.96 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $278,594.41 \text{ W/m}^2$
Kinetic flame height $547.08 \mu\text{m}$
Average kinetic flame density 10.04 kg/m^3
Average kinetic flame temperature 1,602.94 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $7.87\text{e}+07 \text{ W/m}^2$
Kinetic flame height $2.26 \mu\text{m}$
Average kinetic flame density 9.29 kg/m^3
Average kinetic flame temperature 1,731.94 K
Heat flux density due to metal combustion, $4.49\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $0.33 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.22 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $0.11 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-5.5152085010057306\text{E}-08$
Skeleton layer thickness $3.4\text{e}-05 \text{ m}$
Pore diameter $1.13\text{e}-05 \text{ m}$

Diffusion flame heat flux density $401,245.86 \text{ W/m}^2$
Diffusion flame height $706.08 \mu\text{m}$
Heat flux density to surface within skeletal structure $1.94\text{e}+07 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $213,631.7 \text{ W/m}^2$
Total heat flux density to surface $2\text{e}+07 \text{ W/m}^2$
Binder sublimation heat flux density $2\text{e}+07 \text{ W/m}^2$
Surface heat balance error -0.005 W/m^2

At pressure 6.5 MPa

Experimental burning rate 50.5 mm/s
Calculated burning rate 41.18 mm/s

First Conditional Fuel

Surface temperature 900 K
Linear burning rate 112.82 mm/s
Mass decomposition rate $187.17 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $2.47\text{e}+08 \text{ W/m}^2$
Kinetic flame height $1.14 \mu\text{m}$
Average kinetic flame density 11.2 kg/m^3
Average kinetic flame temperature 2,303.5 K
Binder sublimation heat flux density $2.47\text{e}+08 \text{ W/m}^2$
Surface heat balance error -0.044 W/m^2

Secondary Conditional Fuel

Surface temperature 840.97 K
Linear burning rate 19.92 mm/s
Mass decomposition rate $33.05 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $505,328.18 \text{ W/m}^2$
Kinetic flame height $301.59 \mu\text{m}$
Average kinetic flame density 16.09 kg/m^3
Average kinetic flame temperature 1,602.98 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $1\text{e}+08 \text{ W/m}^2$
Kinetic flame height $1.78 \text{ }\mu\text{m}$
Average kinetic flame density 14.9 kg/m^3
Average kinetic flame temperature $1,731.98 \text{ K}$
Heat flux density due to metal combustion, $4.5\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $0.33 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.22 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $0.11 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-5.5152085010057306\text{E}-08$
Skeleton layer thickness $3.4\text{e}-05 \text{ m}$
Pore diameter $1.12\text{e}-05 \text{ m}$
Diffusion flame heat flux density $400,860.52 \text{ W/m}^2$
Diffusion flame height $706.74 \text{ }\mu\text{m}$
Heat flux density to surface within skeletal structure $1.58\text{e}+07 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $428,858.36 \text{ W/m}^2$
Total heat flux density to surface $1.66\text{e}+07 \text{ W/m}^2$
Binder sublimation heat flux density $1.66\text{e}+07 \text{ W/m}^2$
Surface heat balance error -0.0037 W/m^2

Propellant Composition "Bas_4"

At pressure 1 MPa

Experimental burning rate 9.73 mm/s
Calculated burning rate 10.02 mm/s

First Conditional Fuel

Surface temperature 825.11 K
Linear burning rate 11.99 mm/s
Mass decomposition rate $19.88 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $2.4\text{e}+07 \text{ W/m}^2$
Kinetic flame height $12 \text{ }\mu\text{m}$
Average kinetic flame density 1.75 kg/m^3
Average kinetic flame temperature $2,266.05 \text{ K}$
Binder sublimation heat flux density $2.4\text{e}+07 \text{ W/m}^2$
Surface heat balance error $-1.4\text{e}-04 \text{ W/m}^2$

Secondary Conditional Fuel

Surface temperature 788.15 K
Linear burning rate 3.39 mm/s
Mass decomposition rate $5.62 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $5,591.28 \text{ W/m}^2$
Kinetic flame height $4,540.09 \text{ }\mu\text{m}$
Average kinetic flame density 4.34 kg/m^3
Average kinetic flame temperature 915.08 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $5.3\text{e}+06 \text{ W/m}^2$
Kinetic flame height $8.55 \text{ }\mu\text{m}$
Average kinetic flame density 3.91 kg/m^3
Average kinetic flame temperature $1,014.58 \text{ K}$
Heat flux density due to metal combustion, $5.59\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $1.43 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $1.41 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $0.017 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $9.240271880983641\text{E}-10$
Skeleton layer thickness $1.31\text{e}-04 \text{ m}$
Pore diameter $2.84\text{e}-05 \text{ m}$
Diffusion flame heat flux density $29,475.47 \text{ W/m}^2$
Diffusion flame height $9,471.77 \text{ }\mu\text{m}$
Heat flux density to surface within skeletal structure $5.76\text{e}+06 \text{ W/m}^2$

Heat flux density to surface outside skeletal structure 2,634.79 W/m²
Total heat flux density to surface 5.79e+06 W/m²
Binder sublimation heat flux density 5.79e+06 W/m²
Surface heat balance error 0.001 W/m²

At pressure 4.06 MPa

Experimental burning rate 25.92 mm/s
Calculated burning rate 26.14 mm/s

First Conditional Fuel

Surface temperature 879.86 K
Linear burning rate 64.1 mm/s
Mass decomposition rate 106.33 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface 1.37e+08 W/m²
Kinetic flame height 2.06 μm
Average kinetic flame density 7.02 kg/m³
Average kinetic flame temperature 2,293.43 K
Binder sublimation heat flux density 1.37e+08 W/m²
Surface heat balance error -0.015 W/m²

Secondary Conditional Fuel

Surface temperature 812.68 K
Linear burning rate 7.94 mm/s
Mass decomposition rate 13.17 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 13,653.13 W/m²
Kinetic flame height 1,679.62 μm
Average kinetic flame density 17.36 kg/m³
Average kinetic flame temperature 927.34 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 4.71e+06 W/m²
Kinetic flame height 9.1 μm
Average kinetic flame density 15.68 kg/m³
Average kinetic flame temperature 1,026.84 K
Heat flux density due to metal combustion, 1.01e+07 W/m²
Effective thermal conductivity 1.42 W/(m·K)
Conductive thermal conductivity 1.41 W/(m·K)
Radiative thermal conductivity 0.011 W/(m·K)
Conductive thermal conductivity balance error 9.240271880983641E-10
Skeleton layer thickness 6.84e-05 m
Pore diameter 1.82e-05 m
Diffusion flame heat flux density 73,702.36 W/m²
Diffusion flame height 3,754.72 μm
Heat flux density to surface within skeletal structure 7.79e+06 W/m²
Heat flux density to surface outside skeletal structure 6,485.52 W/m²
Total heat flux density to surface 7.87e+06 W/m²
Binder sublimation heat flux density 7.87e+06 W/m²
Surface heat balance error 7.56e-04 W/m²

At pressure 6.5 MPa

Experimental burning rate 36.06 mm/s
Calculated burning rate 36.05 mm/s

First Conditional Fuel

Surface temperature 900 K
Linear burning rate 112.82 mm/s
Mass decomposition rate 187.17 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface 2.47e+08 W/m²
Kinetic flame height 1.14 μm
Average kinetic flame density 11.2 kg/m³
Average kinetic flame temperature 2,303.5 K
Binder sublimation heat flux density 2.47e+08 W/m²

Surface heat balance error -0.044 W/m²

Secondary Conditional Fuel

Surface temperature 821.71 K

Linear burning rate 10.72 mm/s

Mass decomposition rate 17.79 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 18,116.86 W/m²

Kinetic flame height 1,215.96 μm

Average kinetic flame density 27.69 kg/m³

Average kinetic flame temperature 931.85 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 4.46e+06 W/m²

Kinetic flame height 9.4 μm

Average kinetic flame density 25.01 kg/m³

Average kinetic flame temperature 1,031.35 K

Heat flux density due to metal combustion, 1.25e+07 W/m²

Effective thermal conductivity 1.42 W/(m·K)

Conductive thermal conductivity 1.41 W/(m·K)

Radiative thermal conductivity 0.0095 W/(m·K)

Conductive thermal conductivity balance error 9.240271880983641E-10

Skeleton layer thickness 5.44e-05 m

Pore diameter 1.56e-05 m

Diffusion flame heat flux density 55,699.69 W/m²

Diffusion flame height 4,952.08 μm

Heat flux density to surface within skeletal structure 8.84e+06 W/m²

Heat flux density to surface outside skeletal structure 8,664.17 W/m²

Total heat flux density to surface 8.9e+06 W/m²

Binder sublimation heat flux density 8.9e+06 W/m²

Surface heat balance error -1.59e-04 W/m²

Propellant Composition "Bas_1"

At pressure 1 MPa

Experimental burning rate 14.26 mm/s

Calculated burning rate 14.28 mm/s

First Conditional Fuel

Surface temperature 609.54 K

Linear burning rate 11.44 mm/s

Mass decomposition rate 18.98 kg·s⁻¹·m⁻²

Kinetic flame heat flux density to surface 1.68e+07 W/m²

Kinetic flame height 18.45 μm

Average kinetic flame density 1.84 kg/m³

Average kinetic flame temperature 2,158.27 K

Binder sublimation heat flux density 1.68e+07 W/m²

Surface heat balance error 0.0051 W/m²

Secondary Conditional Fuel

Surface temperature 600 K

Linear burning rate 7.46 mm/s

Mass decomposition rate 12.37 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 118,481.6 W/m²

Kinetic flame height 1,489.68 μm

Average kinetic flame density 2.68 kg/m³

Average kinetic flame temperature 1,482.5 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 5.25e+06 W/m²

Kinetic flame height 18.89 μm

Average kinetic flame density 3.62 kg/m³

Average kinetic flame temperature 1,095.5 K

Heat flux density due to metal combustion, $2.09\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $3.1\text{e}-06 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-4.092294297874943\text{E}-08$
Skeleton layer thickness $1.35\text{e}-04 \text{ m}$
Pore diameter $1.93\text{e}-09 \text{ m}$
Diffusion flame heat flux density $298,909.56 \text{ W/m}^2$
Diffusion flame height $1,025.73 \mu\text{m}$
Heat flux density to surface within skeletal structure $2.77\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $73,689.12 \text{ W/m}^2$
Total heat flux density to surface $9.83\text{e}+06 \text{ W/m}^2$
Binder sublimation heat flux density $9.83\text{e}+06 \text{ W/m}^2$
Surface heat balance error $5.2\text{e}-04 \text{ W/m}^2$

At pressure 4.06 MPa

Experimental burning rate 38.01 mm/s
Calculated burning rate 38.1 mm/s

First Conditional Fuel

Surface temperature 649.82 K
Linear burning rate 60.76 mm/s
Mass decomposition rate $100.79 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $9.52\text{e}+07 \text{ W/m}^2$
Kinetic flame height $3.21 \mu\text{m}$
Average kinetic flame density 7.39 kg/m^3
Average kinetic flame temperature $2,178.41 \text{ K}$
Binder sublimation heat flux density $9.52\text{e}+07 \text{ W/m}^2$
Surface heat balance error -0.022 W/m^2

Secondary Conditional Fuel

Surface temperature 619.28 K
Linear burning rate 17.48 mm/s
Mass decomposition rate $28.99 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $300,636.68 \text{ W/m}^2$
Kinetic flame height $580.68 \mu\text{m}$
Average kinetic flame density 10.79 kg/m^3
Average kinetic flame temperature $1,492.14 \text{ K}$

Skeleton Structure Region

Kinetic flame heat flux density to surface $1.85\text{e}+07 \text{ W/m}^2$
Kinetic flame height $5.26 \mu\text{m}$
Average kinetic flame density 14.57 kg/m^3
Average kinetic flame temperature $1,105.14 \text{ K}$
Heat flux density due to metal combustion, $3.89\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $1.99\text{e}-06 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-4.092294297874943\text{E}-08$
Skeleton layer thickness $7.06\text{e}-05 \text{ m}$
Pore diameter $1.23\text{e}-09 \text{ m}$
Diffusion flame heat flux density $195,416.16 \text{ W/m}^2$
Diffusion flame height $1,559.09 \mu\text{m}$
Heat flux density to surface within skeletal structure $4.48\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $240,420.28 \text{ W/m}^2$
Total heat flux density to surface $1.54\text{e}+07 \text{ W/m}^2$
Binder sublimation heat flux density $1.54\text{e}+07 \text{ W/m}^2$
Surface heat balance error $8.04\text{e}-05 \text{ W/m}^2$

At pressure 6.5 MPa

Experimental burning rate 52.88 mm/s

Calculated burning rate 52.53 mm/s

First Conditional Fuel

Surface temperature 664.64 K
Linear burning rate 106.68 mm/s
Mass decomposition rate $176.98 \text{ kg} \cdot \text{s}^{-1} \cdot \text{m}^{-2}$
Kinetic flame heat flux density to surface $1.71\text{e}+08 \text{ W/m}^2$
Kinetic flame height 1.78 μm
Average kinetic flame density 11.8 kg/m^3
Average kinetic flame temperature 2,185.82 K
Binder sublimation heat flux density $1.71\text{e}+08 \text{ W/m}^2$
Surface heat balance error -0.031 W/m^2

Secondary Conditional Fuel

Surface temperature 626.25 K
Linear burning rate 23.47 mm/s
Mass decomposition rate $38.94 \text{ kg} \cdot \text{s}^{-1} \cdot \text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $408,132.6 \text{ W/m}^2$
Kinetic flame height 426.03 μm
Average kinetic flame density 17.25 kg/m^3
Average kinetic flame temperature 1,495.62 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $2.8\text{e}+07 \text{ W/m}^2$
Kinetic flame height 3.44 μm
Average kinetic flame density 23.27 kg/m^3
Average kinetic flame temperature 1,108.62 K
Heat flux density due to metal combustion, $4.82\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $0.4 \text{ W/(m} \cdot \text{K)}$
Conductive thermal conductivity $0.4 \text{ W/(m} \cdot \text{K)}$
Radiative thermal conductivity $1.7\text{e}-06 \text{ W/(m} \cdot \text{K)}$
Conductive thermal conductivity balance error $-4.092294297874943\text{E}-08$
Skeleton layer thickness $5.64\text{e}-05 \text{ m}$
Pore diameter $1.06\text{e}-09 \text{ m}$

Diffusion flame heat flux density $145,828.04 \text{ W/m}^2$
Diffusion flame height 2,084.48 μm
Heat flux density to surface within skeletal structure $5.36\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $341,499.08 \text{ W/m}^2$
Total heat flux density to surface $1.83\text{e}+07 \text{ W/m}^2$
Binder sublimation heat flux density $1.83\text{e}+07 \text{ W/m}^2$
Surface heat balance error -0.0056 W/m^2

Propellant Composition "Bas_0"

At pressure 1 MPa

Experimental burning rate 5.97 mm/s
Calculated burning rate 5.96 mm/s

First Conditional Fuel

Surface temperature 825.11 K
Linear burning rate 11.99 mm/s
Mass decomposition rate $19.88 \text{ kg} \cdot \text{s}^{-1} \cdot \text{m}^{-2}$
Kinetic flame heat flux density to surface $2.4\text{e}+07 \text{ W/m}^2$
Kinetic flame height 12 μm
Average kinetic flame density 1.75 kg/m^3
Average kinetic flame temperature 2,266.05 K
Binder sublimation heat flux density $2.4\text{e}+07 \text{ W/m}^2$
Surface heat balance error -0.0049 W/m^2

Secondary Conditional Fuel

Surface temperature 781.52 K
Linear burning rate 2.67 mm/s
Mass decomposition rate $4.43 \text{ kg} \cdot \text{s}^{-1} \cdot \text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 348,308.26 W/m²

Kinetic flame height 454.62 μm

Average kinetic flame density 2.52 kg/m³

Average kinetic flame temperature 1,573.26 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 6,301.94 W/m²

Kinetic flame height 12,844.89 μm

Average kinetic flame density 3.35 kg/m³

Average kinetic flame temperature 1,186.26 K

Heat flux density due to metal combustion, 1.55e+06 W/m²

Effective thermal conductivity 0.4 W/(m·K)

Conductive thermal conductivity 0.4 W/(m·K)

Radiative thermal conductivity 3.84e-07 W/(m·K)

Conductive thermal conductivity balance error -4.092294297874943E-08

Skeleton layer thickness 1.35e-04 m

Pore diameter 2.39e-10 m

Diffusion flame heat flux density 786,283.68 W/m²

Diffusion flame height 366.85 μm

Heat flux density to surface within skeletal structure 402,456.58 W/m²

Heat flux density to surface outside skeletal structure 258,238.74 W/m²

Total heat flux density to surface 4.52e+06 W/m²

Binder sublimation heat flux density 4.52e+06 W/m²

Surface heat balance error 5.76e-04 W/m²

At pressure 4.06 MPa

Experimental burning rate 13.63 mm/s

Calculated burning rate 13.66 mm/s

First Conditional Fuel

Surface temperature 879.86 K

Linear burning rate 64.1 mm/s

Mass decomposition rate 106.33 kg·s⁻¹·m⁻²

Kinetic flame heat flux density to surface 1.37e+08 W/m²

Kinetic flame height 2.06 μm

Average kinetic flame density 7.02 kg/m³

Average kinetic flame temperature 2,293.43 K

Binder sublimation heat flux density 1.37e+08 W/m²

Surface heat balance error 0.032 W/m²

Secondary Conditional Fuel

Surface temperature 803.44 K

Linear burning rate 5.8 mm/s

Mass decomposition rate 9.62 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 949,964.53 W/m²

Kinetic flame height 164.38 μm

Average kinetic flame density 10.16 kg/m³

Average kinetic flame temperature 1,584.22 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 3,816.49 W/m²

Kinetic flame height 20,635.65 μm

Average kinetic flame density 13.44 kg/m³

Average kinetic flame temperature 1,197.22 K

Heat flux density due to metal combustion, 2.68e+06 W/m²

Effective thermal conductivity 0.4 W/(m·K)

Conductive thermal conductivity 0.4 W/(m·K)

Radiative thermal conductivity 2.56e-07 W/(m·K)

Conductive thermal conductivity balance error -4.092294297874943E-08

Skeleton layer thickness 7.48e-05 m

Pore diameter 1.59e-10 m

Diffusion flame heat flux density 553,621.13 W/m²
Diffusion flame height 517.06 μm
Heat flux density to surface within skeletal structure 503,121.4 W/m²
Heat flux density to surface outside skeletal structure 771,789.98 W/m²
Total heat flux density to surface 5.71e+06 W/m²
Binder sublimation heat flux density 5.71e+06 W/m²
Surface heat balance error 4.47e-04 W/m²

At pressure 6.5 MPa

Experimental burning rate 18 mm/s

Calculated burning rate 17.94 mm/s

First Conditional Fuel

Surface temperature 900 K
Linear burning rate 112.82 mm/s
Mass decomposition rate 187.17 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface 2.47e+08 W/m²
Kinetic flame height 1.14 μm
Average kinetic flame density 11.2 kg/m³
Average kinetic flame temperature 2,303.5 K
Binder sublimation heat flux density 2.47e+08 W/m²
Surface heat balance error -0.029 W/m²

Secondary Conditional Fuel

Surface temperature 811.14 K
Linear burning rate 7.54 mm/s
Mass decomposition rate 12.5 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 1.33e+06 W/m²
Kinetic flame height 116.79 μm
Average kinetic flame density 16.25 kg/m³
Average kinetic flame temperature 1,588.07 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 3,220.01 W/m²
Kinetic flame height 24,219.19 μm
Average kinetic flame density 21.48 kg/m³
Average kinetic flame temperature 1,201.07 K
Heat flux density due to metal combustion, 3.22e+06 W/m²
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 2.23e-07 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 6.13e-05 m
Pore diameter 1.39e-10 m

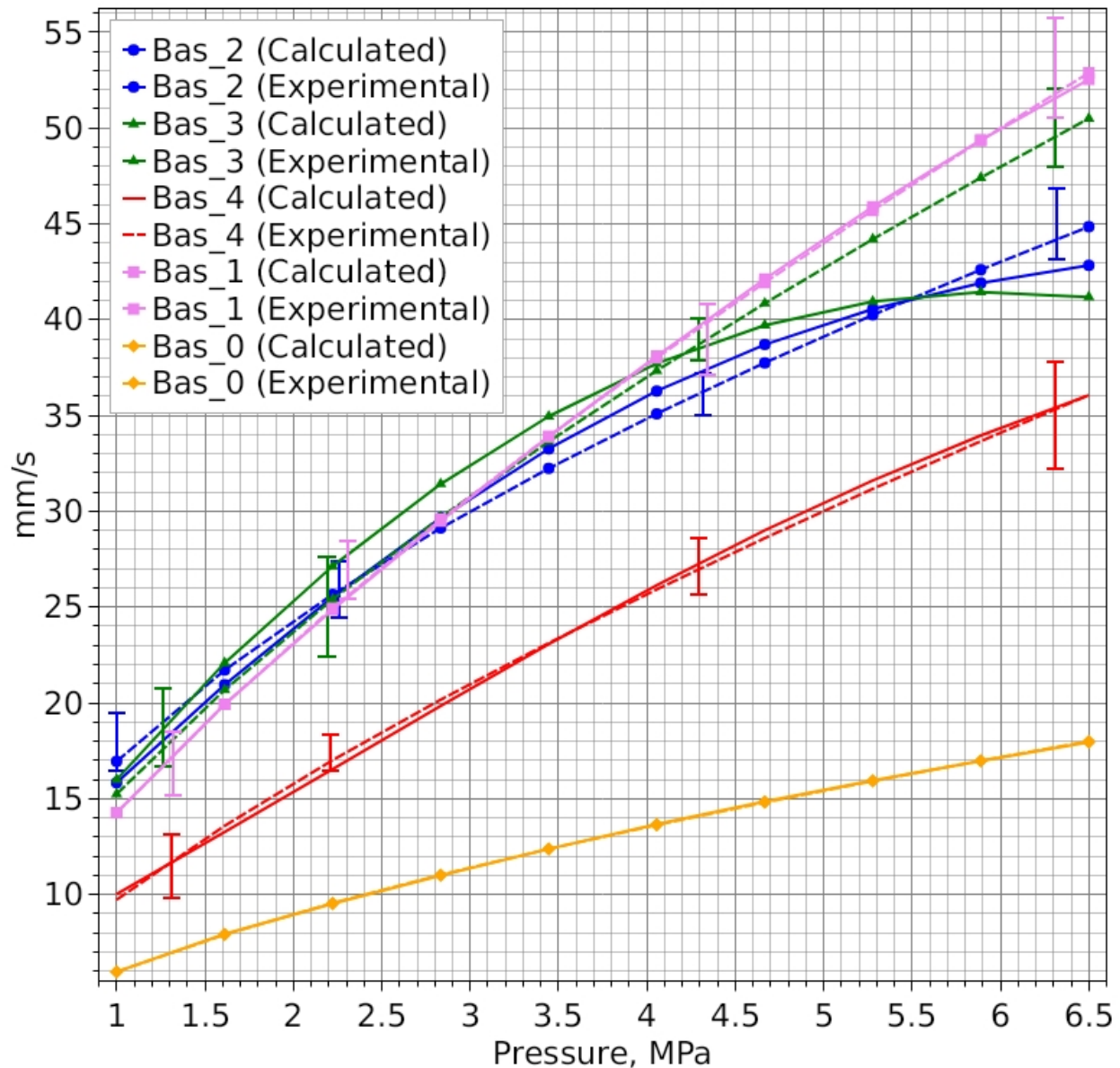
Diffusion flame heat flux density 426,589.31 W/m²
Diffusion flame height 669.23 μm
Heat flux density to surface within skeletal structure 406,390.25 W/m²
Heat flux density to surface outside skeletal structure 1.16e+06 W/m²
Total heat flux density to surface 6.24e+06 W/m²
Binder sublimation heat flux density 6.24e+06 W/m²
Surface heat balance error -0.0012 W/m²

1 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	16.94 mm/s		15.24 mm/s		9.73 mm/s		14.26 mm/s		5.97 mm/s	
Calculated Burning Rate	15.83 mm/s		16 mm/s		10.02 mm/s		14.28 mm/s		5.96 mm/s	
Difference	1.12 mm/s		-0.76 mm/s		-0.29 mm/s		-0.012 mm/s		0.0023 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	11.99 mm/s	8.41 mm/s	11.99 mm/s	11.99 mm/s	11.99 mm/s	3.39 mm/s	11.44 mm/s	7.46 mm/s	11.99 mm/s	2.67 mm/s
Surface Temperature	825.11 K	814.38 K	825.11 K	825.11 K	825.11 K	788.15 K	609.54 K	600 K	825.11 K	781.52 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	2.4e+07 W/m ²		2.4e+07 W/m ²		2.4e+07 W/m ²		1.68e+07 W/m ²		2.4e+07 W/m ²	
Kinetic Flame Heat Flux Density (Skeleton)	9.14e+06 W/m ²		6.32e+07 W/m ²		5.3e+06 W/m ²		5.25e+06 W/m ²		6,301.94 W/m ²	
Kinetic Flame Height (Skeleton)	8.5 μm		2.84 μm		8.55 μm		18.89 μm		12,844.89 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	111,119.8 W/m ²		78,048.98 W/m ²		5,591.28 W/m ²		118,481.6 W/m ²		348,308.26 W/m ²	
Kinetic Flame Height (Out-Skeleton)	1,395.44 μm		1,972.98 μm		4,540.09 μm		1,489.68 μm		454.62 μm	
Diffusion Flame Heat Flux Density	241,299.64 W/m ²		478,243.65 W/m ²		29,475.47 W/m ²		298,909.56 W/m ²		786,283.68 W/m ²	
Diffusion Flame Height	1,156.08 μm		595.7 μm		9,471.77 μm		1,025.73 μm		366.85 μm	
Metal Combustion Heat Flux Density	3.2e+06 W/m ²		3.49e+06 W/m ²		5.59e+06 W/m ²		2.09e+06 W/m ²		1.55e+06 W/m ²	

4.06 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	35.09 mm/s		37.34 mm/s		25.92 mm/s		38.01 mm/s		13.63 mm/s	
Calculated Burning Rate	36.28 mm/s		37.71 mm/s		26.14 mm/s		38.1 mm/s		13.66 mm/s	
Difference	-1.19 mm/s		-0.37 mm/s		-0.22 mm/s		-0.093 mm/s		-0.037 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	64.1 mm/s	16.44 mm/s	64.1 mm/s	19.87 mm/s	64.1 mm/s	7.94 mm/s	60.76 mm/s	17.48 mm/s	64.1 mm/s	5.8 mm/s
Surface Temperature	879.86 K	834.9 K	879.86 K	840.88 K	879.86 K	812.68 K	649.82 K	619.28 K	879.86 K	803.44 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	1.37e+08 W/m ²		1.37e+08 W/m ²		1.37e+08 W/m ²		9.52e+07 W/m ²		1.37e+08 W/m ²	
Kinetic Flame Heat Flux Density (Skeleton)	9.73e+06 W/m ²		7.87e+07 W/m ²		4.71e+06 W/m ²		1.85e+07 W/m ²		3,816.49 W/m ²	
Kinetic Flame Height (Skeleton)	7.77 μm		2.26 μm		9.1 μm		5.26 μm		20,635.65 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	336,371.76 W/m ²		278,594.41 W/m ²		13,653.13 W/m ²		300,636.68 W/m ²		949,964.53 W/m ²	
Kinetic Flame Height (Out-Skeleton)	454.88 μm		547.08 μm		1,679.62 μm		580.68 μm		164.38 μm	
Diffusion Flame Heat Flux Density	188,791.26 W/m ²		401,245.86 W/m ²		73,702.36 W/m ²		195,416.16 W/m ²		553,621.13 W/m ²	
Diffusion Flame Height	1,466.75 μm		706.08 μm		3,754.72 μm		1,559.09 μm		517.06 μm	
Metal Combustion Heat Flux Density	5.01e+06 W/m ²		4.49e+06 W/m ²		1.01e+07 W/m ²		3.89e+06 W/m ²		2.68e+06 W/m ²	

6.5 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	44.85 mm/s		50.5 mm/s		36.06 mm/s		52.88 mm/s		18 mm/s	
Calculated Burning Rate	42.84 mm/s		41.18 mm/s		36.05 mm/s		52.53 mm/s		17.94 mm/s	
Difference	2.01 mm/s		9.32 mm/s		0.0095 mm/s		0.35 mm/s		0.058 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	112.82 mm/s	18.74 mm/s	112.82 mm/s	19.92 mm/s	112.82 mm/s	10.72 mm/s	106.68 mm/s	23.47 mm/s	112.82 mm/s	7.54 mm/s
Surface Temperature	900 K	839.02 K	900 K	840.97 K	900 K	821.71 K	664.64 K	626.25 K	900 K	811.14 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	2.47e+08 W/m ²		2.47e+08 W/m ²		2.47e+08 W/m ²		1.71e+08 W/m ²		2.47e+08 W/m ²	
Kinetic Flame Heat Flux Density (Skeleton)	1.09e+07 W/m ²		1e+08 W/m ²		4.46e+06 W/m ²		2.8e+07 W/m ²		3,220.01 W/m ²	
Kinetic Flame Height (Skeleton)	6.9 μm		1.78 μm		9.4 μm		3.44 μm		24,219.19 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	537,172.4 W/m ²		505,328.18 W/m ²		18,116.86 W/m ²		408,132.6 W/m ²		1.33e+06 W/m ²	
Kinetic Flame Height (Out-Skeleton)	284.08 μm		301.59 μm		1,215.96 μm		426.03 μm		116.79 μm	
Diffusion Flame Heat Flux Density	166,181.49 W/m ²		400,860.52 W/m ²		55,699.69 W/m ²		145,828.04 W/m ²		426,589.31 W/m ²	
Diffusion Flame Height	1,663.83 μm		706.74 μm		4,952.08 μm		2,084.48 μm		669.23 μm	
Metal Combustion Heat Flux Density	5.47e+06 W/m ²		4.5e+06 W/m ²		1.25e+07 W/m ²		4.82e+06 W/m ²		3.22e+06 W/m ²	

Burning Rates - Group Optimization (All Propellants)



Propellant Composition Parameters

Composition "Bas_2"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.

Vieille's Power Law: $U = A \cdot p^v$, $A = 1.285341E-05$; $v = 0.52$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Diffusion flame temperature 3,604 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,365 K

Composition "Bas_3"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.57.

Vieille's Power Law: $U = A \cdot p^v$, $A = 2.203002E-06$; $v = 0.64$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,674 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,623 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,365 K

Composition "Bas_4"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.56.

Vieille's Power Law: $U = A \cdot p^v$, $A = 6.137894E-07$; $v = 0.7$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,580 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,241 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,042 K

Composition "Bas_1"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.

Vieille's Power Law: $U = A \cdot p^v$, $A = 9.000448E-07$; $v = 0.7$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Diffusion flame temperature 3,666 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 2,365 K

Composition "Bas_0"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.

Vieille's Power Law: $U = A \cdot p^v$, $A = 1.720582E-06$; $v = 0.59$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Diffusion flame temperature 3,666 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

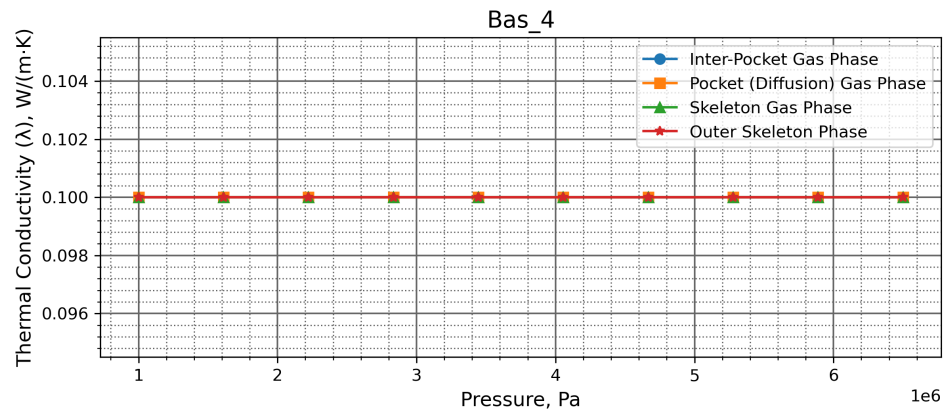
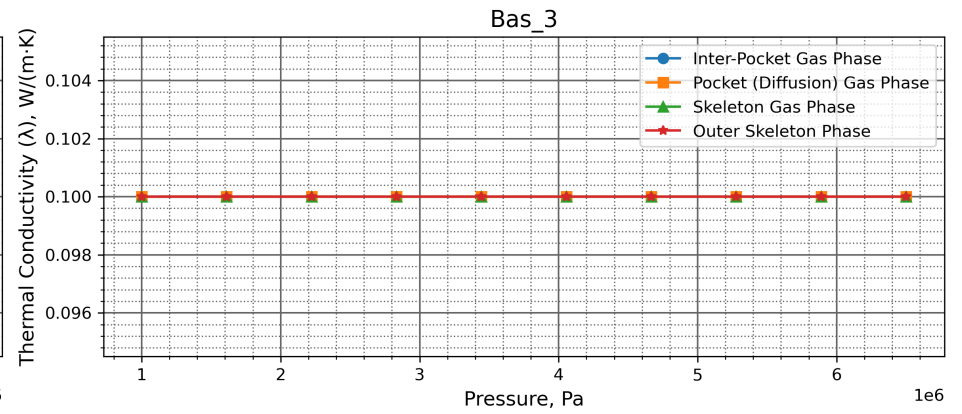
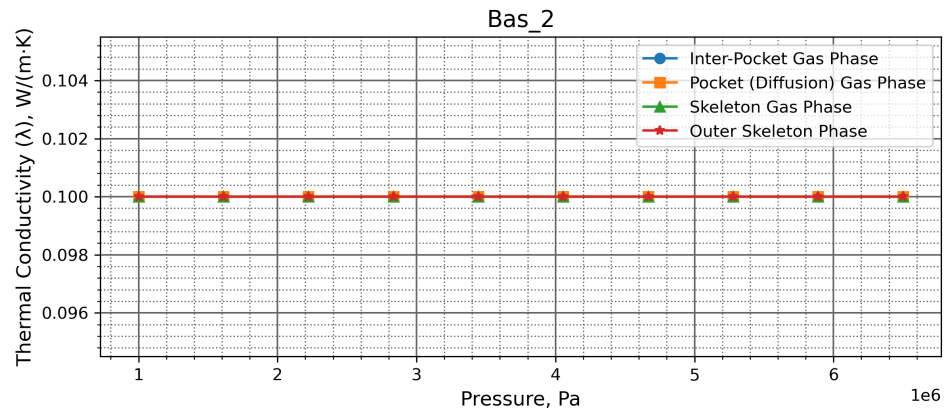
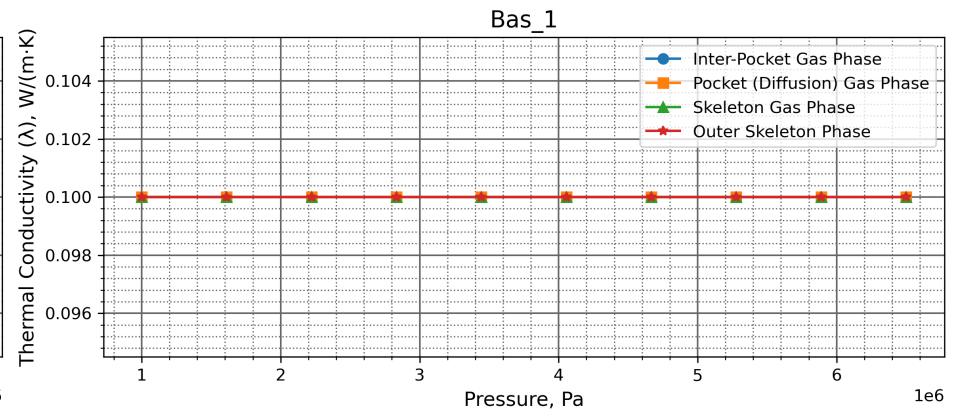
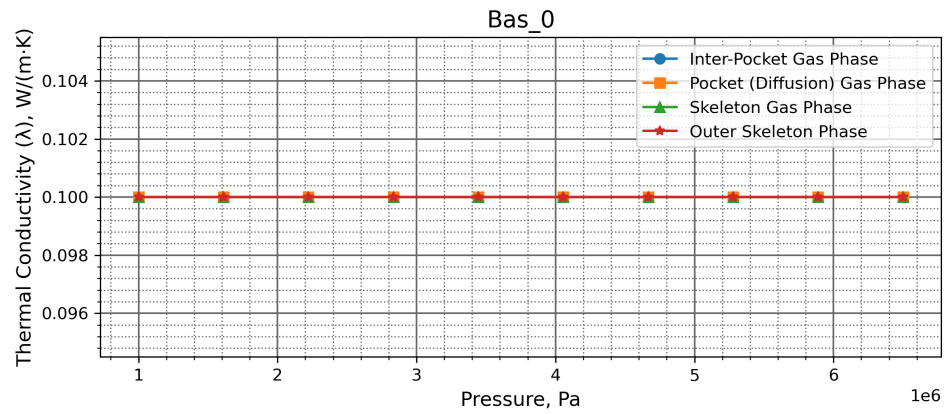
Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

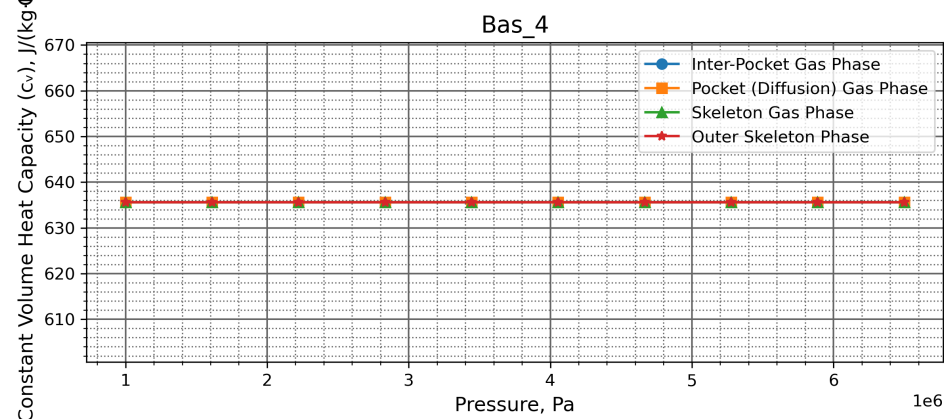
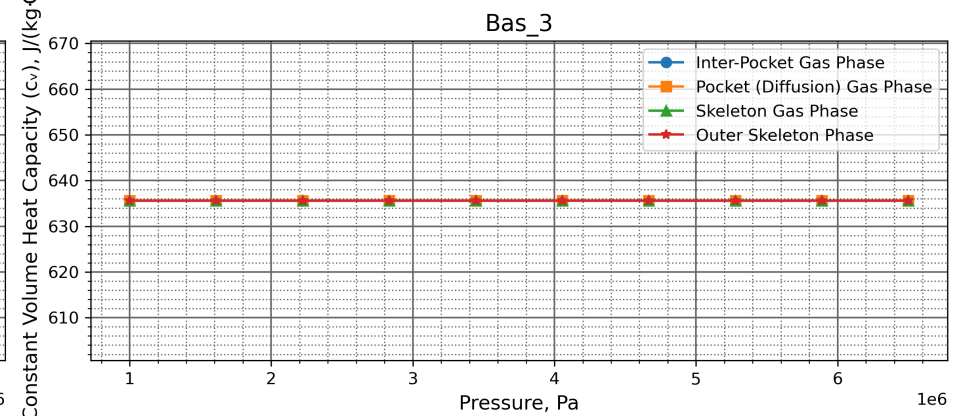
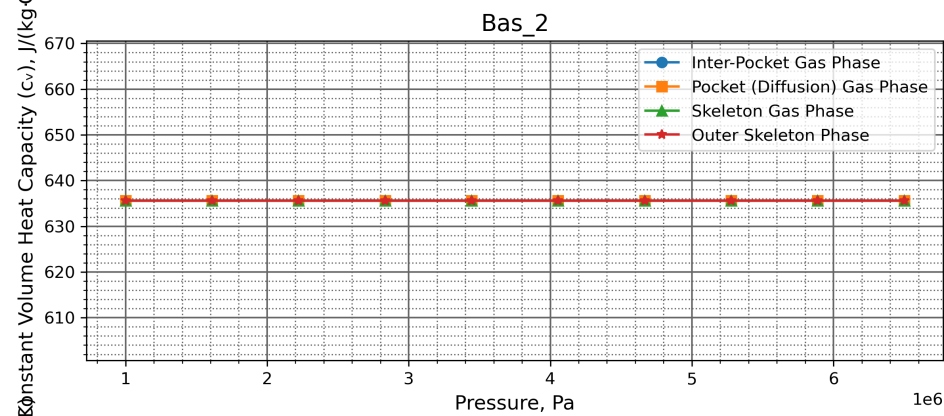
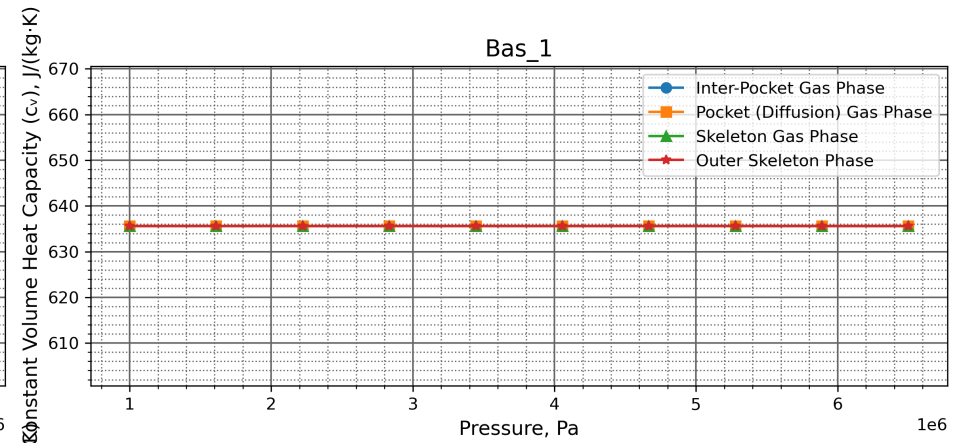
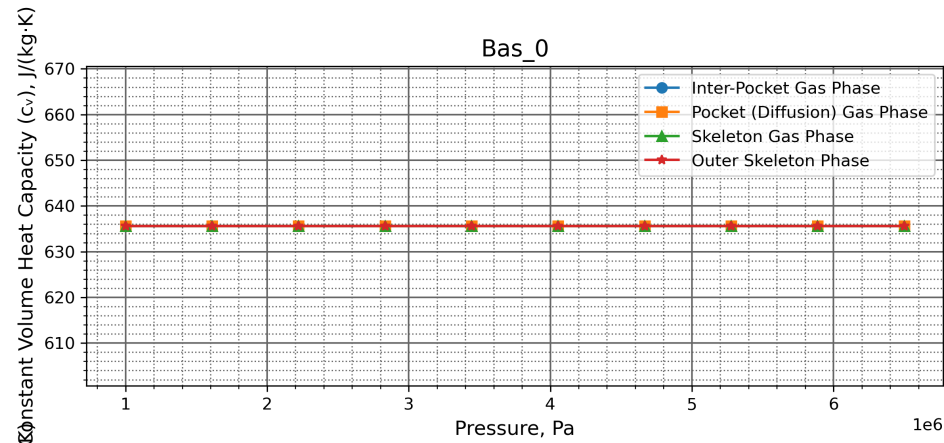
Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 2,365 K

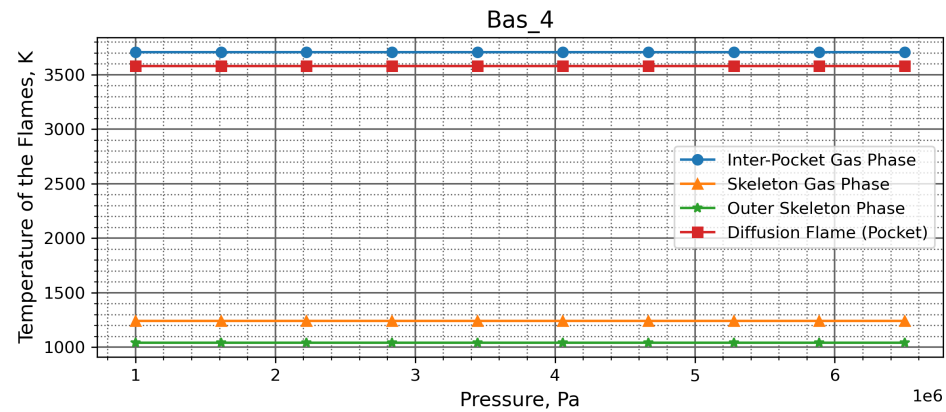
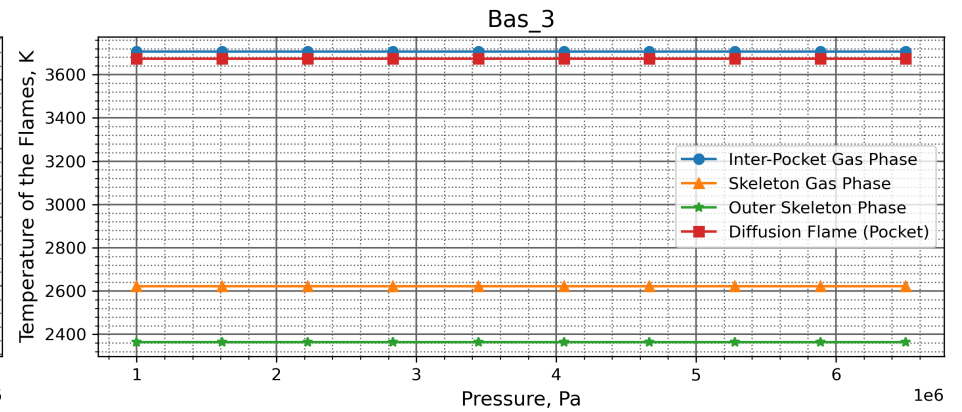
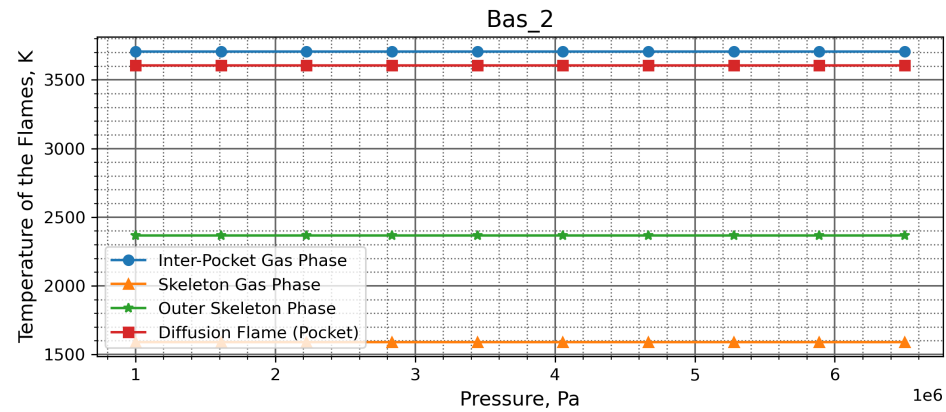
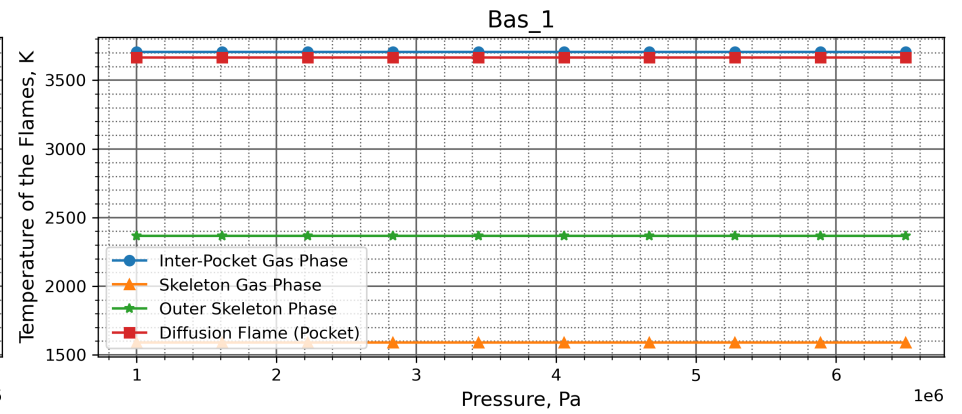
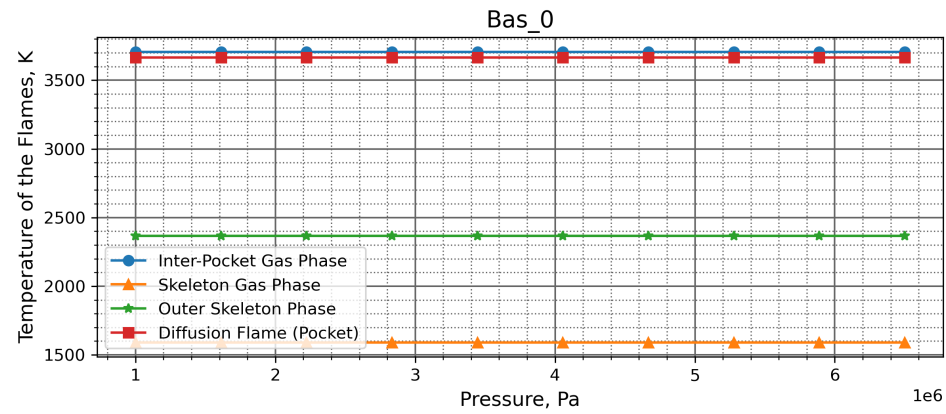
Thermal Conductivity (λ), W/(m·K)



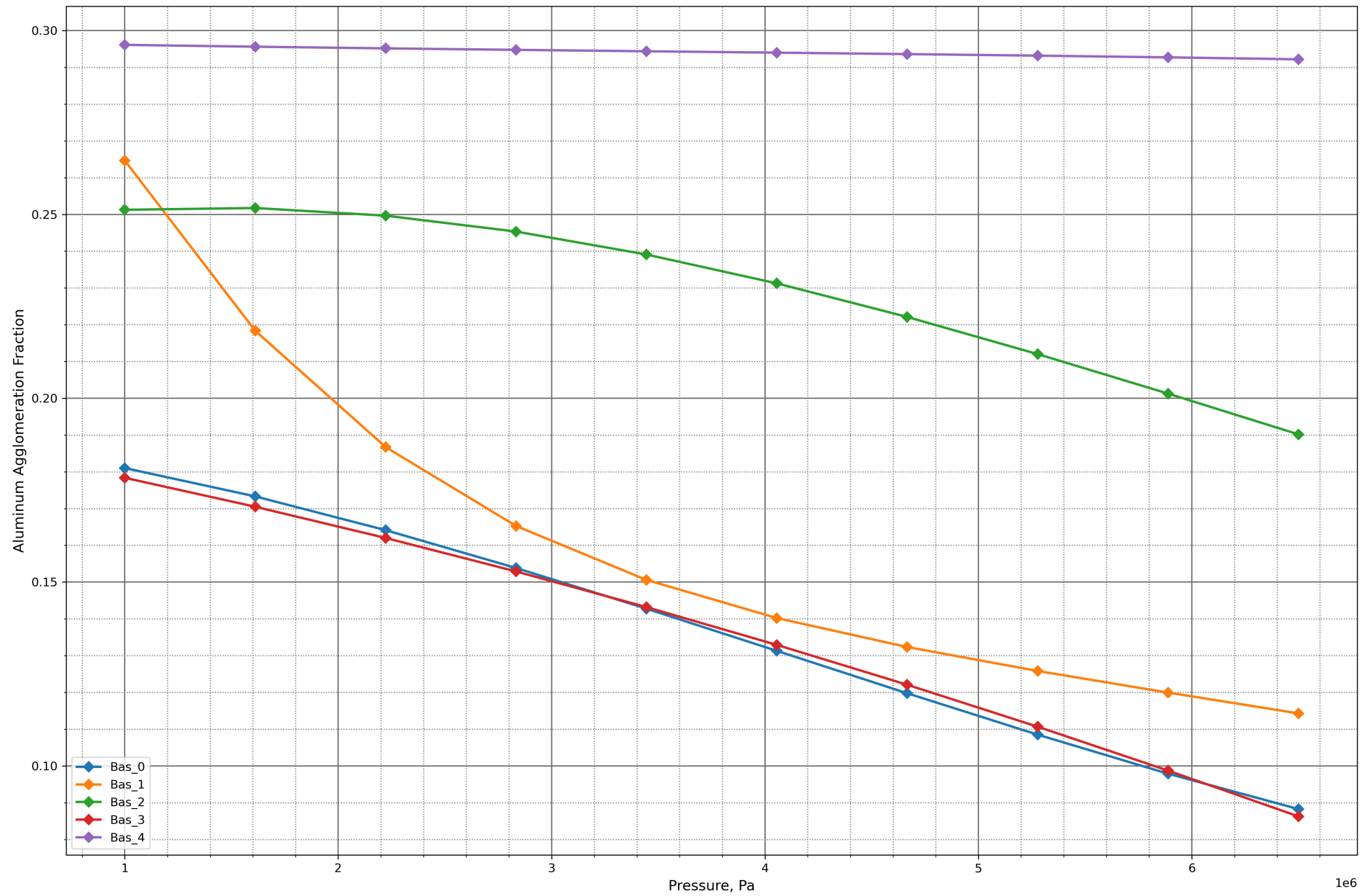
Constant Volume Heat Capacity (c_v), J/(kg·K)



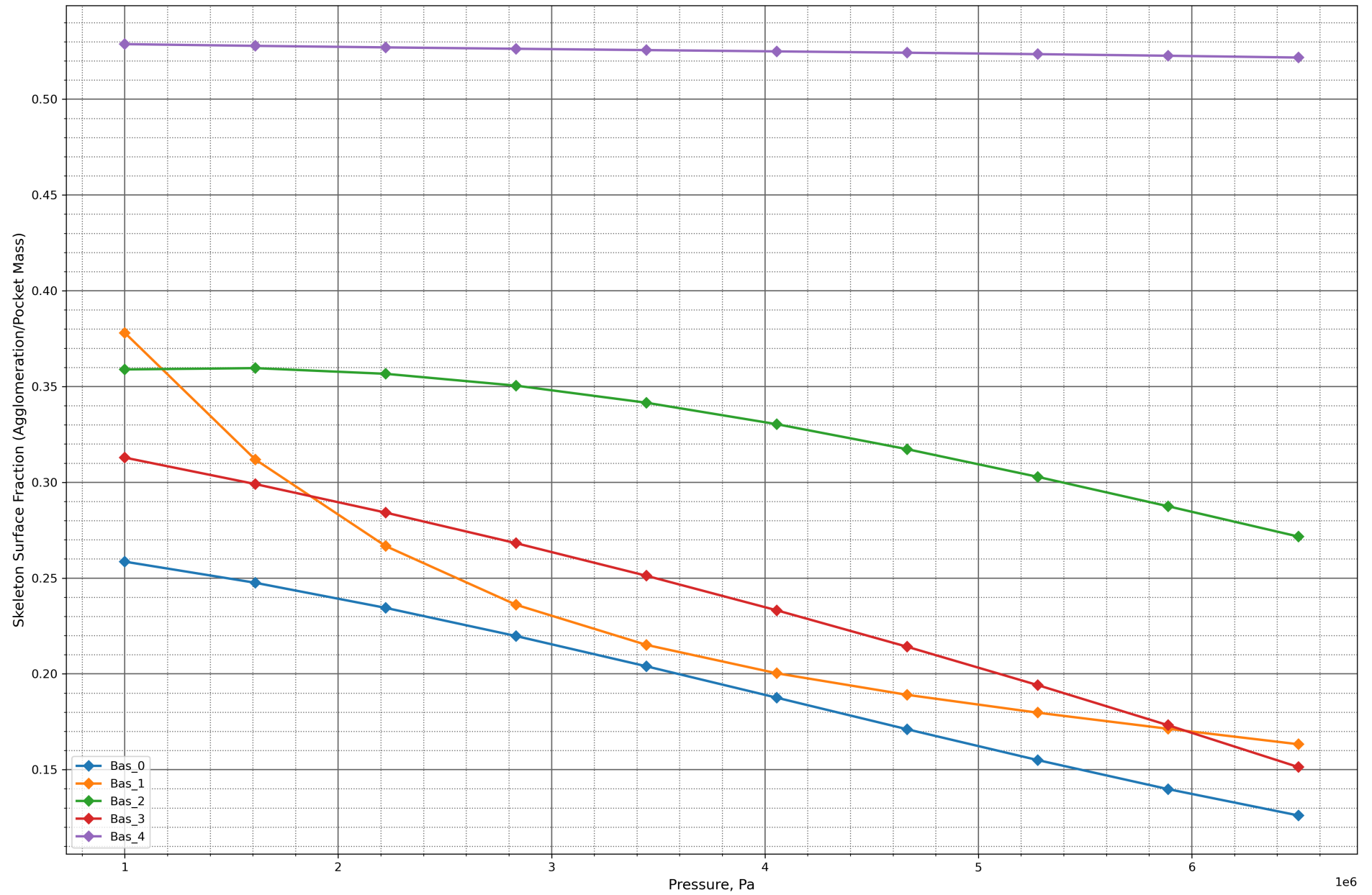
Temperature of the Flames, K



Aluminum Agglomeration Fraction



Skeleton Surface Fraction (Agglomeration/Pocket Mass)



Performance Report

System Information

Operating System: Unix 6.8.0.124 (Ubuntu 24.04.4 LTS)
Processor: Intel(R) Xeon(R) CPU E5-2697 v4 @ 2.30GHz (Base Clock Frequency 2.29 GHz)
Memory: 15.6 GB (64-bit), Unknown
Machine Name: vscode
Runtime: .NET 10.0.9 (.NET 10.0.9)
Architecture: X64 / X64

Total Execution Time

Total execution time: 09:01:33.902

Garbage Collection Information

Total Memory: 8704920 bytes
GC Pause Time Percentage: 0.06%
Generation 0 Collections: 25375
Generation 1 Collections: 2239
Generation 2 Collections: 249

Detailed Performance Metrics

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance

Call count: 31104844

Maximum execution time: 8.6156 ms

Minimum execution time: 0.0001 ms

Mean execution time: 0.6103800376430821 ms

Execution time standard deviation: 0.19898383292899144 ms

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance

Call count: 25446617

Maximum execution time: 9.6377 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.23037628570430418 ms

Execution time standard deviation: 0.06432811958181309 ms

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance

Call count: 23109941

Maximum execution time: 8.0183 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2142852116888012 ms

Execution time standard deviation: 0.08630760806478364 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance

Call count: 31004400

Maximum execution time: 11.7095 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.6072275076343358 ms

Execution time standard deviation: 0.20140359481199818 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance

Call count: 25212753

Maximum execution time: 4.9587 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.23121287502002486 ms

Execution time standard deviation: 0.06520901757679036 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance

Call count: 22819799

Maximum execution time: 14.2066 ms

Minimum execution time: 0.0002 ms
Mean execution time: 0.2129988774090191 ms
Execution time standard deviation: 0.08706414024898519 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 30118560
Maximum execution time: 7.6179 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.608213893466347 ms
Execution time standard deviation: 0.20052072854028902 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 24517253
Maximum execution time: 5.3373 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.23040587866836085 ms
Execution time standard deviation: 0.06501516368421503 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 22176666
Maximum execution time: 6.3207 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.21409070641637362 ms
Execution time standard deviation: 0.08628343785313168 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 30118560
Maximum execution time: 9.5244 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.6119113249935382 ms
Execution time standard deviation: 0.20225472816652287 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 24552357
Maximum execution time: 4.8767 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.23028047889250025 ms
Execution time standard deviation: 0.06469064094438494 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 22203492
Maximum execution time: 4.6289 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.21306486636881766 ms
Execution time standard deviation: 0.087126989832275 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance
Call count: 30118560
Maximum execution time: 15.9357 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.6103296227076019 ms
Execution time standard deviation: 0.20178602428299397 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance
Call count: 24522364
Maximum execution time: 6.3961 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.23174059958085164 ms
Execution time standard deviation: 0.06495477420282258 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance

Call count: 22209607

Maximum execution time: 7.9216 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.21463341835363475 ms

Execution time standard deviation: 0.08636403102698585 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance

Call count: 30118560

Maximum execution time: 18.7526 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.6087079846116229 ms

Execution time standard deviation: 0.20058696231650977 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance

Call count: 24531324

Maximum execution time: 16.0119 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.23038055990371925 ms

Execution time standard deviation: 0.06462020643621467 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance

Call count: 22228320

Maximum execution time: 5.7295 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.21463582420531924 ms

Execution time standard deviation: 0.08604502708223172 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance

Call count: 30118560

Maximum execution time: 9.9375 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.6082397461931255 ms

Execution time standard deviation: 0.20144138740182224 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance

Call count: 24499492

Maximum execution time: 8.2163 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2308478167629386 ms

Execution time standard deviation: 0.06480541775657535 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance

Call count: 22202009

Maximum execution time: 10.5673 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.21421640393894073 ms

Execution time standard deviation: 0.08576764956409208 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance

Call count: 30118560

Maximum execution time: 13.455 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.6097876986316414 ms

Execution time standard deviation: 0.19982756199184756 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance

Call count: 24585606

Maximum execution time: 4.5831 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2305148221850241 ms
Execution time standard deviation: 0.0642247227269354 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance

Call count: 22261216
Maximum execution time: 6.662 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2134775677528134 ms
Execution time standard deviation: 0.08650307414299253 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance

Call count: 30118560
Maximum execution time: 6.3401 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.6110009114247261 ms
Execution time standard deviation: 0.201102871965249 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance

Call count: 24536913
Maximum execution time: 6.9169 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.23036274931567713 ms
Execution time standard deviation: 0.06442363736771824 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance

Call count: 22212658
Maximum execution time: 7.9847 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2149545318394491 ms
Execution time standard deviation: 0.08594066400384452 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance

Call count: 30118560
Maximum execution time: 21.0882 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.6078943213786733 ms
Execution time standard deviation: 0.20149890811829635 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance

Call count: 24481602
Maximum execution time: 7.9253 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.23012319542241794 ms
Execution time standard deviation: 0.06488640343352094 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance

Call count: 22123267
Maximum execution time: 34.2096 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.21335544625937633 ms
Execution time standard deviation: 0.08731976283886891 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance

Call count: 30118560
Maximum execution time: 8.4196 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.6109961905683914 ms

Execution time standard deviation: 0.19924619362706478 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance

Call count: 24622482

Maximum execution time: 7.8256 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.23157514524733802 ms

Execution time standard deviation: 0.0643020272362375 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance

Call count: 22340871

Maximum execution time: 7.7764 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2157248463141522 ms

Execution time standard deviation: 0.08654858445795177 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance

Call count: 30118560

Maximum execution time: 15.7823 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.6083098237399752 ms

Execution time standard deviation: 0.20218493123627287 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance

Call count: 24466345

Maximum execution time: 6.9689 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.229839006022366 ms

Execution time standard deviation: 0.06518029464917567 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance

Call count: 22108142

Maximum execution time: 5.1896 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.21373949361276579 ms

Execution time standard deviation: 0.08719183937227472 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance

Call count: 30118560

Maximum execution time: 8.5103 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.6097707569717755 ms

Execution time standard deviation: 0.20144962732839813 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance

Call count: 24509341

Maximum execution time: 6.4113 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.23026759122567142 ms

Execution time standard deviation: 0.06435258140334295 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance

Call count: 22195529

Maximum execution time: 4.3777 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.21422800087351002 ms

Execution time standard deviation: 0.08607825423914824 ms